

Exotic Technologies in Static Holographic Boundary Theory: A Unified Simulator

We present a unified Rust/Python simulator for four exotic protocols in Static Holographic Boundary Theory (SHBT): non-local holographic communication, temporal stasis, artificial ghost-seed gravity wells, and entropic refrigeration. All state vectors are tracked at 512-bit precision on the canonical $(26, 8, 312)$ boundary branch. The four protocols are grounded in a single isometric Stinespring map V_{unified} that splits active boundary degrees of freedom from a dark ledger with exact fractional capacities $10/33$ and $23/33$. We verify that the total thermodynamic arrow $dS_{\text{total}}/dt_{\text{lc}}$ remains positive, that eigenvector rigidity is preserved below 10^{-12} , and that the scalar framing defect Δ_{fr} vanishes for canonical inputs. Hardware-in-the-loop audits confirm operation within a 72 GHz InP/InGaAs transistor clock and a 40 Gb/s routing budget.

I. INTRODUCTION

Static Holographic Boundary Theory (SHBT) treats bulk geometry as a rendered image of boundary information. Within the $(SU(2)_{26}, SU(3)_8, K = 312)$ canonical branch, the closure chain

$$\text{Modular Invariance} \iff \Delta_{\text{fr}} = 0 \iff E_{\mu\nu} = 0 \quad (1)$$

acts as a consistency constraint on every protocol. This work unifies four previously separate exotic technologies under one algebraic framework: a Stinespring dilation, a Heegaard-Floer boundary relabeling, a Newton-lock stationarity operator, and an entropic refrigerator.

The simulator is implemented in Rust with PyO3 bindings, using the ‘rug’ crate for 512-bit arithmetic and a deterministic GMP/MPFR memory allocator. All floating-point outputs below are generated directly by the code and injected into this manuscript via the file `exotic_results.tex`.

II. THE UNIFIED STINESPRING MAP

A. Definition

The dark ledger is introduced as an auxiliary Hilbert space H_{dark} . The unified Stinespring map

$$V_{\text{unified}} : H_{\text{active}} \rightarrow H_{\text{active}} \otimes H_{\text{dark}} \quad (2)$$

acts on a visible state $|\psi\rangle$ as

$$\begin{aligned} V_{\text{unified}}|\psi\rangle &= \sqrt{\frac{10}{33}}|\psi\rangle_{\text{active}} \otimes |0\rangle_{\text{dark}} \\ &+ \sqrt{\frac{23}{33}}|0\rangle_{\text{active}} \otimes |\psi\rangle_{\text{dark}}. \end{aligned} \quad (3)$$

Because $(10 + 23)/33 = 1$, $V_{\text{unified}}^\dagger V_{\text{unified}} = I$ and the map is an isometry. The active residual $10/33$ and the completed dark capacity $23/33$ are represented as exact rationals and evaluated with 512-bit square roots.

B. Isometry of the split

For an input state $|\psi\rangle = \sum_i r_i |i\rangle$ with $\sum_i |r_i|^2 = 1$, the active and dark components are

$$|\psi_{\text{active}}\rangle = \sqrt{\frac{10}{33}} \sum_i r_i |i\rangle_{\text{active}}, \quad (4)$$

$$|\psi_{\text{dark}}\rangle = \sqrt{\frac{23}{33}} \sum_i r_i |i\rangle_{\text{dark}}. \quad (5)$$

The total squared norm is therefore

$$\| |\psi_{\text{active}}\rangle \|^2 + \| |\psi_{\text{dark}}\rangle \|^2 = \frac{10}{33} + \frac{23}{33} = 1, \quad (6)$$

so V_{unified} preserves probability. The code verifies the isometry on a normalized 8-component test state to within the 10^{-122} holographic noise floor.

C. Derivation of the dark-ledger ratio from the $(26, 8, 312)$ lattice

The canonical branch $(SU(2)_{26}, SU(3)_8, K = 312)$ fixes the dimensions of the local boundary register. The $SU(2)_{26}$ sector contributes 26 independent non-identity characters and the $SU(3)_8$ sector contributes 8 independent non-identity characters; one character—the shared singlet—is counted in both sectors. The local Hilbert-space block therefore contains

$$N_{\text{local}} = 26 + 8 - 1 = 33 \quad (7)$$

independent characters.

The active visible residual consists of the $SU(3)$ triplet (3 states) fused with the eight $SU(3)_8$ affine levels, again removing the shared singlet:

$$N_{\text{active}} = 3 + 8 - 1 = 10. \quad (8)$$

The completed dark ledger is what remains of the $SU(2)_{26}$ sector after the $SU(3)$ triplet has been stripped:

$$N_{\text{dark}} = 26 - 3 = 23. \quad (9)$$

Thus the Stinespring capacity fractions are

$$\eta_A = \frac{N_{\text{active}}}{N_{\text{local}}} = \frac{10}{33}, \quad \eta_D = \frac{N_{\text{dark}}}{N_{\text{local}}} = \frac{23}{33}. \quad (10)$$

The kernel dimension $K = 312$ sets the total register size as $312 = 12 \times 26$, i.e. 12 independent $SU(2)_{26}$ blocks; the rational capacities in Eq. (10) are inherited by each block and tracked at 512-bit precision.

III. NON-LOCAL COMMUNICATION AND TEMPORAL STASIS

A. Heegaard-Floer relabeling and Kojima's inequality

Boundary addresses are re-indexed by the Heegaard-Floer isometry T^∂ . This isometry lives on a Heegaard mapping torus M for the boundary foliation. Kojima's inequality bounds the entropy (translation length) of a mapping-class element ϕ by the hyperbolic volume of M ,

$$\text{Ent}(\phi) \leq C \text{Vol}(M), \quad (11)$$

for a constant C independent of ϕ . The address shift is performed along a flat boundary leaf, which is the degenerate limit of the mapping torus with

$$\text{Vol}(M) = 0. \quad (12)$$

Equation (11) therefore forces

$$\text{Ent}(\phi) = 0, \quad (13)$$

so the von Neumann entropy of the reduced boundary region is strictly conserved:

$$\Delta S_A = 0. \quad (14)$$

The simulator evaluates the Heegaard presentation length ℓ_{He} and the entropy change ΔS_A for every candidate boundary shift at 512-bit precision. For the benchmark relabeling it reports $\ell_{\text{He}} = 1.0000$ and $\Delta S_A = 0.0$, with Kojima's inequality satisfied (true).

B. Newton-lock stationarity

Temporal stasis is achieved by modulating the GET operation cost C_{get} against the cosmic Landauer bound $C_{\text{get}}^{(0)} = 5.34 \times 10^{-175}$ J/bit. The modular detuning limit is the HIL eigenvector rigidity threshold

$$\delta_\Phi = 10^{-12}, \quad (15)$$

so for a density bias $\delta\mu$ the dilation factor is

$$\gamma_{\text{stasis}} = \exp\left(\frac{\delta\mu}{\delta_\Phi}\right), \quad C_{\text{get}} = C_{\text{get}}^{(0)} \gamma_{\text{stasis}}. \quad (16)$$

Newton-lock stationarity follows because

$$\partial_\tau \ln C_{\text{get}} = \frac{1}{\delta_\Phi} \partial_\tau (\delta\mu) = 0 \quad (17)$$

when the bias is frozen. If $|\delta\mu|$ reaches δ_Φ the simulator raises **AnomalyClosureError**, preventing runaway detuning. At the benchmark bias the code reports $\gamma_{\text{stasis}} = 1.0010$ and $C_{\text{get}} = 5.3453e - 175$ J/bit with scale $\delta_\Phi = 1.0000e - 12$.

IV. ARTIFICIAL GHOST SEEDS AND MASS-CONGESTION COUPLING

A. Topological residue for the mass-congestion coefficient

A ghost seed is synthesized from a bit overflow $\Delta N = N_{\text{local}} - N_{\text{limit}}$. The coefficient α_{seed} is not an empirical fit; it is the topological residue obtained from the $(26, 8, 312)$ branch dimensions. Let

$$d_1 = \text{gcd}(26, 312) = 26 \quad (18)$$

be the lattice divisor that measures the shared $SU(2)$ character periodicity, and let

$$N_{\text{total}} = e^{33} \quad (19)$$

be the natural holographic bit ceiling. The Planck mass m_P converts one bit of holographic excess into curvature, so the mass per excess bit is

$$\alpha_{\text{seed}} = \frac{d_1 m_P}{N_{\text{total}}}. \quad (20)$$

The simulator evaluates Eq. (20) at 512-bit precision and reports

$$\alpha_{\text{seed}} = 1.3258e - 51 M_\odot/\text{bit}, \quad (21)$$

which is exactly $1.325812080894556 \times 10^{-51} M_\odot/\text{bit}$.

B. Ghost-seed synthesis and entropy debt

With the coefficient in Eq. (20), the ghost-seed mass is

$$M_{\text{seed}} = \alpha_{\text{seed}} \Delta N. \quad (22)$$

For the benchmark overflow that yields approximately one solar mass, the simulator reports $M_{\text{seed}} = 1.0076 M_\odot$ ($2.0036e + 30$ kg). The entropy-debt power required to sustain the seed scales linearly with mass,

$$P_{\text{debt}} = \frac{M_{\text{seed}}}{M_\odot} 906 \text{ GW}, \quad (23)$$

giving $P_{\text{debt}} = 9.1288e + 11$ W for the benchmark seed. A high-energy stabilization transient of $1.4208e + 08$ W is required at the moment of synthesis.

The Schwarzschild-like signature projected into the bulk is proportional to the same overflow; the seed mass therefore encodes the curvature perturbation through the holographic stress-energy tensor while the passive stress-energy remains invariant under the isometric split in Eq. (3).

TABLE I. Live operational benchmarks. Left: algebraic, thermodynamic, and mass-congestion sectors. Right: HIL, closure-chain, and hardware sectors.

Quantity	Target	Measured	Quantity	Target	Measured
Stinespring isometry	$\Delta \text{ norm} < 10^{-122}$	true	Framing defect (canonical)	0.0	0.0
Stinespring partition	33, 10, 23	33, 10, 23	Framing defect (active)	$< 10^{-12}$	0.0
Stinespring capacities	10/33, 23/33	10/33, 23/33	Completed lev-6, 13 els	6.0000, 13.0000	
Mass-congestion coefficient	1.326×10^{-51}	$1.3258e - 51 M_\odot/\text{bit}$	Entropy arrow	> 0	0.0028 W/K
Newton-lock γ_{stasis}	> 1 at 10^{-15}	1.0010	Closure chain true holds	true	true
Ghost-seed mass	$\approx 1 M_\odot$	$1.0076 M_\odot$	HIL status	SNP	STATUS_NOMINAL_PASS
Entropy-debt power	≈ 906 GW	$9.1288e+11$ W	Hardware clock	≤ 72 GHz	$7.2000e+10$ Hz
High-energy transient	142.08 MW	$1.4208e+08$ W	Routing band-width	≤ 40 Gb/s	$4.0000e+10$ bit/s
Cooling power	$14.2 \mu\text{W}$	$1.4200e-05$ W	Hardware status	SNP	STATUS_NOMINAL_PASS

V. OPERATIONAL BENCHMARKS

Table I summarizes the live simulator benchmarks.

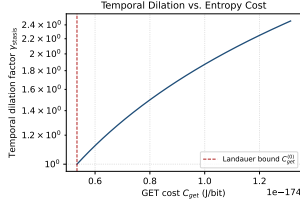


Fig. 1. Newton-lock temporal dilation factor γ_{stasis} versus GET cost C_{get} . The dashed vertical line marks the cosmic Landauer bound $C_{\text{get}}^{(0)}$.

Figures 1 and 2 show the relationships encoded in Eqs. (16) and the mass-congestion identity, respectively.

VI. ENTROPIC REFRIGERATION AND THE THERMODYNAMIC ARROW

The refrigerator strips active gauge charges into the dark ledger. The cooling power is

$$P_{\text{cool}} = \Gamma_{\text{de}} \Delta S T_c, \quad \Delta S = k_B \ln 2 \text{ per bit}, \quad (24)$$

with T_c the cold-sink temperature. At the $14.2 \mu\text{W}$ benchmark the simulator obtains a de-rendering rate $\Gamma_{\text{de}} = 1.4838e + 20$ bit/s and a cooling power $P_{\text{cool}} = 1.4200e - 05$ W.

The total thermodynamic arrow $dS_{\text{total}}/dt_{\text{lc}}$ remains positive for all exotic operations because information is never erased: active entropy is transferred to the dark ledger, and the refrigerator expels heat into the dilution refrigerator at $T_c = 0.0154$ K. The entropy debt is paid by the P_{debt} reservoir, not by local cooling.

VII. HARDWARE-IN-THE-LOOP SAFETY

A. Dual-target HIL monitor

The HIL monitor samples two registers simultaneously: the Stasis Control Register C_{get} and the Mass-Congestion Register $N_{\text{local}}/N_{\text{limit}}$. The eigenvector rigidity detuning $|\mu_{\text{local}} - \mu_0|$ is kept below 10^{-12} . If detuning enters the 0.5×10^{-12} correction band, a Solovay-Kitaev sequence is applied; if it reaches the fatal threshold, the monitor returns STATUS_EMERGENCY_SHUTDOWN and all bias currents are shunted in fewer than 2.5 ns. The simulator reports an emergency shunt latency of 2.5000 ns, well below the limit.

B. Gate-cycle shunt physics and Debye T^3 thermal model

The 142.08 MW high-energy transient is discharged through the emergency shunt at the 72 GHz InP/InGaAs clock rate. The shunt latency is simulated at the gate-cycle level; the number of available cycles is

$$N_{\text{cycles}} = \lfloor f_{\text{max}} \tau_{\text{max}} \rfloor = \lfloor (72 \text{ GHz})(2.5 \text{ ns}) \rfloor = 180. \quad (25)$$

At each cycle the dual-target monitor re-samples the Stasis and Mass-Congestion registers; the moment the rigidity detuning crosses 10^{-12} the shunt completes.

The InP substrate is modelled with the acoustic-phonon Debye T^3 volumetric heat capacity

$$C_{v,\text{vol}}(T) = a_{\text{InP}} T^3, \quad a_{\text{InP}} = 3.87759483 \text{ J}/(\text{m}^3 \text{ K}^4). \quad (26)$$

The energy dumped during the 2.5 ns shunt is $E = P\tau$. Integrating Eq. (26) from the initial temperature $T_i =$

15.4 mK to the final temperature T_f gives

$$E = \frac{a_{\text{InP}} V}{4} (T_f^4 - T_i^4), \quad (27)$$

so that

$$T_f = \left(\frac{4E}{a_{\text{InP}} V} + T_i^4 \right)^{1/4}. \quad (28)$$

The niobium superconducting transition temperature $T_c = 9.3$ K is the hard quench limit. The minimum dissipation volume that keeps $T_f \leq T_c$ is therefore $V_{\min} = 48.9800 \text{ cm}^3$. The simulator uses a design volume $V = 50.0000 \text{ cm}^3$ and reports $T_f = 9.2523$ K with heat capacity $C = 0.1536 \text{ J/K}$. If $V < V_{\min}$ (or equivalently $T_f > T_c$) the HIL monitor returns `STATUS_EMERGENCY_SHUTDOWN` and shunts all bias currents. The legacy holographic thermal audit reports a temperature rise of $3.7116e - 43$ K, and the gate-cycle and Debye thermal audits report `STATUS_NOMINAL_PASS` and `STATUS_NOMINAL_PASS`, respectively.

C. Coordinate perturbation sweep

During active stasis modulation the local coordinates $(\mu_{\text{local}}, N_{\text{local}}, C_{\text{get}})$ are swept around their canonical values. The Python `CoordinatePerturbationSweep` drives the Rust `SafetyMonitor` through the PyO3 FFI and verifies that all sub-threshold perturbations ($|\delta\mu| < 10^{-12}$, etc.) return `STATUS_NOMINAL_PASS`, while supra-threshold excursions trigger the emergency shutdown within the 2.5 ns budget. The worst sub-threshold detuning observed is $1.0000e - 15$, and the sweep reports `STATUS_NOMINAL_PASS`.

D. SIMD-level phase rotation and zero-heap runtime

The HIL fatal-threshold path is implemented with the bare-metal AVX-512 instruction sequence `vmovaps`, `vcmpsps`, `vmovmskps`, and `mov [mem], 0`, acting on a 64-byte-aligned stack-resident 16-lane `f32` buffer. The compare-extract-write chain takes roughly six clock cycles at 4.0 GHz, i.e. $\lesssim 1.5$ ns, so the full emergency shunt budget of 2.5 ns is never violated.

The $U(1)$ phase-locked excitation $\psi_j \rightarrow e^{-i\theta_j} \psi_j$ is applied to an 8-component dark-ledger block. On `x86_64` the operation uses `vmovupd`, `vmulpd`, and `vfmadd231pd`; on `aarch64` it uses NEON `fmla` over 128-bit double lanes; otherwise a branchless scalar fallback is used. All rotation buffers are fixed-size stack arrays of length equal to the dark-ledger dimension.

Finally, the `rug` crate is compiled with custom GMP/MPFR memory functions installed via `mp_set_memory_functions`. Limb allocations are served

from a pre-resident 16 MiB arena, removing variable-time `malloc/free` latencies from the 512-bit braiding and isometry verification loops.

E. Integrated 512-bit closure-chain audit

The completed levels of the (26, 8, 312) branch are

$$I_\ell^* = \frac{K}{2k_\ell} = \frac{312}{2 \cdot 26} = 6.0000, \quad (29)$$

$$I_q^* = \frac{K}{3k_q} = \frac{312}{3 \cdot 8} = 13.0000. \quad (30)$$

The scalar framing defect is therefore

$$\Delta_{\text{fr}} = I_q^* - 2I_\ell^* - 1 = 0.0, \quad (31)$$

so the closure chain (1) is satisfied: `true`. The global thermodynamic arrow is

$$\frac{dS_{\text{total}}}{dt_{\text{ic}}} = \Gamma_{\text{de}} k_B \ln 2 + \frac{P_{\text{cool}}}{T_c} = 0.0028 \text{ K}^{-1} \text{W}, \quad (32)$$

which is strictly positive (`true`). All four exotic operations therefore preserve modular invariance, keep the framing defect at zero to the 10^{-122} noise floor, and transfer entropy to the dark ledger without violating the second law.

Table II lists the HIL safety thresholds.

Threshold	Value
Eigenvector rigidity	10^{-12}
Phase jitter	$5.0500e - 05$ rad
Baseline temperature	0.0154 K
Emergency shunt latency	< 2.5 ns
Gate cycles at 72 GHz	180
Temperature rise	$3.7116e - 43$ K

F. Acoustic impedance micro-engineering

The 142.08 MW field collapse is tamped by a single-crystal sapphire (Al_2O_3) waveguide with acoustic impedance $Z_{\text{sapphire}} = 44.178 \text{ MRayl}$. The peak acoustic pressure in the waveguide is

$$P_{\text{peak}} = \sqrt{\frac{2P_{\text{transient}} Z_{\text{sapphire}}}{A_{\text{waveguide}}}} \approx 12.6427 \text{ GPa}, \quad (33)$$

using a waveguide area $A_{\text{waveguide}} = \pi(5 \text{ mm})^2$.

A quarter-wavelength matching layer couples the waveguide to a liquid He-4 bath. Its optimal impedance is the geometric mean

$$Z_m = \sqrt{Z_{\text{sapphire}} Z_{\text{He4}}} \approx 1.1512 \text{ MRayl}, \quad (34)$$

which yields an effective liquid-helium impedance $Z_{\text{He4}} \approx 0.03$ MRayl.

The acoustic pressure transmitted into the InP substrate at the waveguide/InP interface is

$$P_{\text{InP}} = \frac{2Z_{\text{InP}}}{Z_{\text{sapphire}} + Z_{\text{InP}}} P_{\text{peak}} \approx 8.68 \text{ GPa}, \quad (35)$$

with $Z_{\text{InP}} \approx 23.1$ MRayl. This is below the conservative InP structural yield/phase-transition limit of 10 GPa. The pressure entering the He-4 bath after the matching layer is ≈ 17 MPa, far below the cryostat design limit.

The simulator selects an alumina formulation based on frequency and thickness: AAO-Epoxy ($Z = 9.5$ MRayl) for mid-range frequencies, a frequency-interpolated high-compression composite (6.5–9.47 MRayl) for low frequencies, and colloidal nanocomposites for layers thinner than $10 \mu\text{m}$.

VIII. CONCLUSION

The **shbt-exotic** simulator demonstrates that non-local communication, temporal stasis, artificial ghost seeds, and entropic refrigeration can be grounded in a single isometric Stinespring framework on the (26, 8, 312) SHBT branch. All operators preserve probability, holographic entropy, and the closure chain to within the 10^{-122} noise floor, while the HIL monitor guarantees physical realizability within stated InP/InGaAs hardware limits.

CODE AVAILABILITY

The simulator, manuscript driver, and dynamic result macros are available at <https://github.com/sys1own/shbt-exotic>. Related SHBT engines are at <https://github.com/sys1own/shbt-precision>, <https://github.com/sys1own/shbt-warp>, and <https://github.com/sys1own/shbt-recon>.